```
/*
Purpose:
--------
This script calculates the Addition, Subtraction, Multiplication and
Division of two numbers.
*/
```

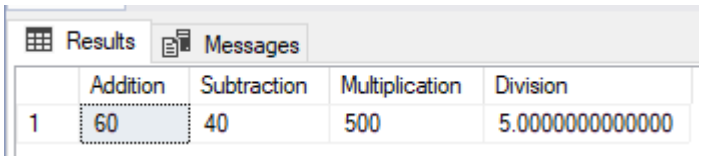
```
Declare    @Number1 int = 50,
           @Number2 int = 10,
           @Addition int,
           @Subtraction int,
           @Multiplication int,
           @Division Decimal(10, 2);

set @Addition = @Number1 + @Number2;
set @Subtraction = @Number1 - @Number2;
set @Multiplication = @Number1 * @Number2;

if @Number2 <> 0
     set @Division = @Number1 / @Number2;
else
     set @Division = NULL;

print 'Addition = ' + CAST(@Addition as nvarchar(10));
print 'Subtraction = ' + CAST(@Subtraction as nvarchar(10));
print 'Multiplication = ' + CAST(@Multiplication as nvarchar(10));
print 'Division = ' + ISNULL(CAST(@Division as nvarchar(50)),
'Cannot divide by zero');
```

----- Query

```
select
     @Number1 + @Number2 AS Addition,
     @Number1 - @Number2 AS Subtraction,
     @Number1 * @Number2 AS Multiplication,
```

```
    CAST(@Number1 AS DECIMAL(10,2)) / NULLIF(@Number2, 0) AS Division;
```

----- Output



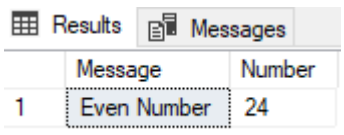
	Addition	Subtraction	Multiplication	Division
1	60	40	500	5.0000000000000

```
/*
Purpose:
--------
This script find Number is Even or Odd.
*/
Declare @Number int = 7;
     if (@Number % 2 = 0)
     begin
          print 'Even Number: ' + cast(@Number as nvarchar(20))
     end

     else
     begin
          print 'Odd Number: ' + cast(@Number as nvarchar(20))
     end
```

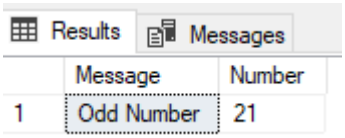
------ Query

```
     select case
               when (@Number % 2 = 0)
               then 'Even Number'
               else 'Odd Number'
               end as Message,
     @Number as Number;
```

----- Output



	Message	Number
1	Even Number	24



	Message	Number
1	Odd Number	21

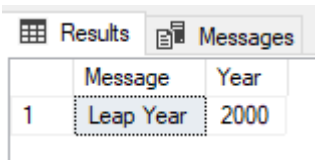
```
/*
Purpose:
--------
This script find year is Leap year or not.
*/
```

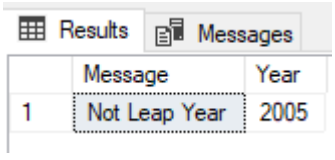
```
Declare @Year int = 2005;
      if (@Year % 4 = 0)
      begin
            print 'Leap Year: ' + cast(@Year as nvarchar(20))
      end

      else
      begin
            print 'Not Leap Year: ' + cast(@Year as nvarchar(20))
      end
```

------ Query

```
      select case
                        when (@Year % 4 = 0)
                        then 'Leap Year'
                        else 'Not Leap Year'
                        end as Message,
      @Year as Year;
```

----- Output





/*
Purpose:

This script Swap 2 numbers with/without using 3rd Variable.

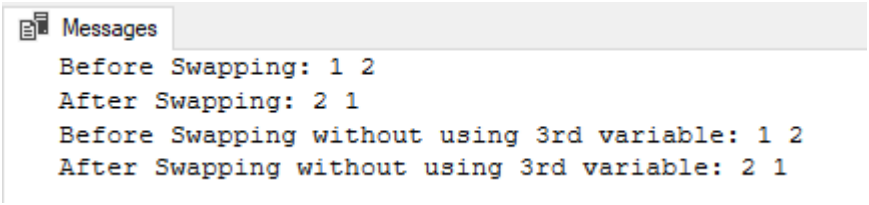
*/

```
Declare @a int = 1,
           @b int = 2,
           @temp int;


     print 'Before Swapping: ' + cast(@a as varchar(5)) + ' ' +
cast(@b as varchar(5));


     set @temp = @a;
     set @a = @b;
     set @b = @temp;


     print 'After Swapping: ' + cast(@a as varchar(5)) + ' ' +
cast(@b as varchar(5));
```

------- Without using 3rd Variable

```
Declare @a1 int = 1,
           @b1 int = 2,
           @temp1 int;


     print 'Before Swapping without using 3rd variable: ' +
cast(@a1 as varchar(5)) + ' ' + cast(@b1 as varchar(5));


     set @a1 = @a1 + @b1;
     set @b1 = @a1 - @b1;
     set @a1 = @a1 - @b1;


     print 'After Swapping without using 3rd variable: ' + cast(@a1
as varchar(5)) + ' ' + cast(@b1 as varchar(5));
```

------ Output

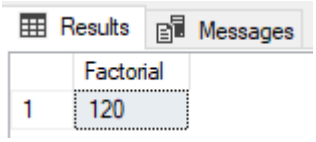



```
/*
Purpose:
--------
This script calculates the factorial of a given number.

Logic:
------
1. Validates whether the input number is negative or not and Returns
an appropriate message.
3. Handles the special case of 0! = 1.
4. Uses a WHILE loop to calculate the factorial for positive
numbers.
5. Displays the calculated factorial value.
*/
```

```
Declare @n int = 0,
           @Factorial bigint = 1,
           @i int = 1;


     if (@n < 0)
begin
    print 'Factorial is not defined for negative numbers.';
end
else if ( @n = 0)
begin
     print 'Factorial = 1';
end
else
begin
     while (@i <= @n)
```

```
	begin
		set @Factorial = @Factorial * @i;
		set @i = @i + 1;
	end

	print 'Factorial = ' + cast(@Factorial as nvarchar(10));
end
```

------ Output




```
/*
Purpose:
--------
This script checks whether the given input is a palindrome.

Logic:
------
1. Accept input as NVARCHAR.
2. Reverse the input using REVERSE().
3. Compare the original and reversed values.
4. Display whether the input is a palindrome or not.

Examples:
---------
121 -> Palindrome
123 -> Not Palindrome
eye -> Palindrome
madam -> Palindrome
hello -> Not Palindrome
*/
```

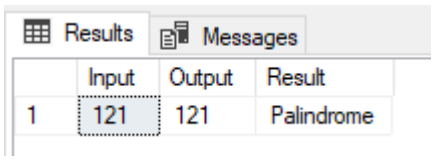
```
Declare @Input nvarchar(10) = '121',
		@Reversed nvarchar(10);

		set @Reversed = REVERSE(@Input);
		if @Input = @Reversed
		begin
```

```
                print 'Palindrom = ' + @Input;
          end
          else
          begin
                print 'Not Palindrom = ' + @Input;
          end
```

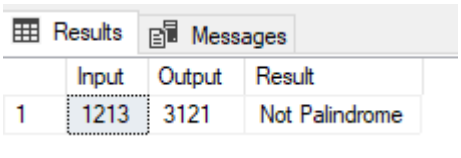
----- Query

```
          select  @Input as Input,
                    @Reversed as Output,
                    case
                          when @Input = @Reversed
                                then 'Palindrome'
                          else
                                'Not Palindrome'
                    end as Result;
```

------ Output



	Input	Output	Result
1	121	121	Palindrome



	Input	Output	Result
1	1213	3121	Not Palindrome

/*

Purpose:

This script calculates the power of a number using a WHILE loop.

Formula:

----

x^y = x × x × x ... (y times)

Examples:

2^3 = 8

5^2 = 25

10^0 = 1

Logic:

---

1. Accept the base number (@x).
2. Accept the exponent value (@y).
3. Initialize the result variable to 1.
4. Multiply the result by the base number repeatedly
   until the exponent count is reached.
5. Display the calculated power value.

Input:

@x : Base number

@y : Exponent value

*/

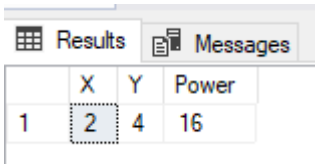
```
Declare @x int = 2,
          @y int = 3,
          @i int = 1,
          @p int = 1;

          while ( @i <= @y )
          begin
               set @p = @p * @x;
               set @i = @i + 1;
```

```
            end
            print 'Power = ' + cast( @p as nvarchar(10) );
```

----- Query

```
            select @x as x, @y as y, @p as Power;
```

----- Output



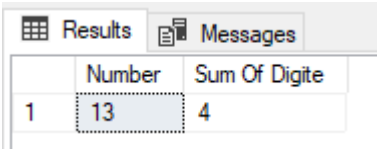

```
/*
Purpose:
--------
This script calculates the Sum of a digit using a WHILE loop.

Examples:
---------
123 = 1 + 2 + 3 = 6

Logic:
------
1. Accept the Number (@n).
2. Display the calculated Sum of value.

Input:
------
@n : Number
*/
Declare @n int = 13,
            @Sum int = 0,
            @k int = 0;
```

```
            while ( @n > 0)
            begin
                  set @k = @n % 10;
                  set @Sum = @Sum + @k;
                  set @n = @n / 10;
            end

            print 'Sum Of Digite = ' + cast ( @Sum as nvarchar(10) );
```

------- Query

```
            select @Sum as [Sum Of Digite];
```

------- Output



/*

Purpose:

This script checks whether a given number is an Armstrong Number.

Examples:

----

153 = 1^3 + 5^3 + 3^3 = 153

370 = 3^3 + 7^3 + 0^3 = 370

371 = 3^3 + 7^3 + 1^3 = 371

407 = 4^3 + 0^3 + 7^3 = 407

Logic:

------

1. Store the original number.

2. Extract each digit using the modulus (%) operator.

3. Calculate the cube of each digit.

4. Add the cubes to a running total.

5. Remove the processed digit using integer division.

6. Compare the calculated sum with the original number.

7. Display whether the number is an Armstrong Number.

Input:

@n : Number to be checked.

*/

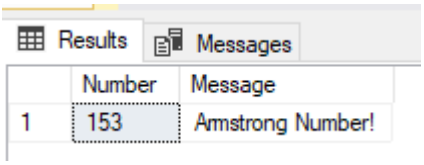
```
Declare @n int = 153,
           @Sum int = 0,
           @k int = 1,
           @p int;

           set @p = @n;

           while ( @n > 0 )
           begin
                set @k = @n % 10;
                set @k = @k * @k *@k;
                set @Sum = @Sum + @k;
                set @n = @n / 10;
           end
```

------- Query

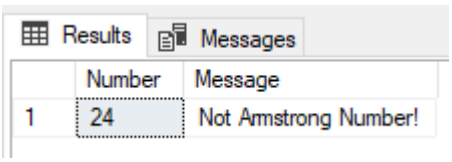
```
           select @Sum as [Number],
                   case
                           when ( @Sum = @p )
                                 then 'Armstrong Number!'
                           else
                                 'Not Armstrong Number!'
```

```
end as Message;
```

-------- Output



	Number	Message
1	153	Armstrong Number!



	Number	Message
1	24	Not Armstrong Number!

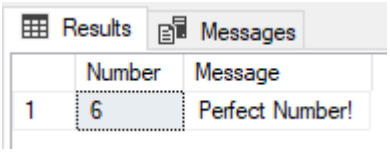
```
/*
Purpose:
--------
This script checks whether a given number is a Perfect Number.
Examples:
---------
6  = 1 + 2 + 3 = 6
28 = 1 + 2 + 4 + 7 + 14 = 28

Logic:
------
1. Accept the input number.
2. Find all divisors of the number excluding itself.
3. Add all the divisors to calculate their sum.
4. Compare the sum of divisors with the original number.
5. Display whether the number is a Perfect Number or not.

Input:
------
@n : Number to be checked.
*/

Declare @n int = 6,
        @Sum int = 0,
        @k int = 1,
```

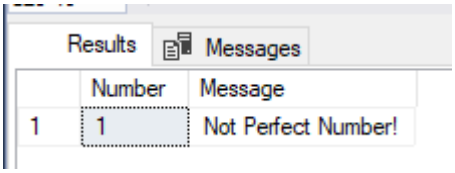
```
            @i int = 1;

            while ( @i < @n)
            begin
                  set @k = @n % @i;

                        if ( @k = 0 )
                        begin
                              set @Sum = @Sum + @i;
                        end
                  set @i = @i + 1;
            end
```

------- Query

```
            select @Sum as [Number],
                     case
                              when ( @Sum = @n )
                                    then 'Perfect Number!'
                              else
                                    'Not Perfect Number!'
                              end as Message;
```

------- Output





/*

Purpose:

This script checks whether a given number is a Prime Number.

Definition:

A Prime Number is a number greater than 1 that has exactly two factors: 1 and itself.

Examples:

2, 3, 5, 7, 11, 13

Logic:

---

1. Accept the input number.
2. Find the count of factors.
3. If the number has exactly two factors, it is Prime.
4. Otherwise, it is Not Prime.

Input:

@n : Number to be checked.

*/

```
Declare @n int = 15,
          @Sum int = 0,
          @i int = 1,
          @Factors nvarchar(10) = '';

          while ( @i <= @n)
          begin
               if ( @n % @i = 0 )
               begin
```

```
                    set @Factors = @Factors + cast(@i as
nvarchar(10)) + ', ';
                    set @Sum = @Sum + 1;
                end
            set @i = @i + 1;
            end
```

------- Query

```
            select @n as [Number], @Factors as Factors, @Sum as
[Factor Counter],
                    case
                            when ( @Sum = 2 )
                                  then 'Prime Number!'
                            else
                                  'Not Prime Number!'
                            end as Message;
```

------- Output

Results Messages

	Number	Factors	Factor Counter	Message
1	7	1, 7,	2	Prime Number!

Results Messages

	Number	Factors	Factor Counter	Message
1	4	1, 2, 4,	3	Not Prime Number!

/*

Purpose:

This script generates the Fibonacci Series up to a specified number of terms.

Examples:

0, 1, 1, 2, 3, 5, 8, 13, 21...

Logic:

------

1. Initialize the first two numbers.

2. Calculate the next number as the sum of the previous two.

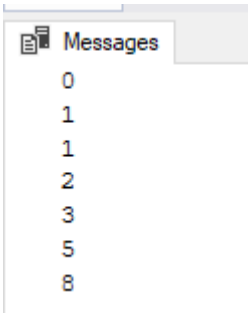
3. Repeat until the required number of terms is generated.

*/

```
Declare @n int = 7,
          @i int = 1,
          @t1 int = 0,
          @t2 int = 1,
          @nextTerm int;

          while ( @i <= @n )
          begin
               if ( @i = 1 )
               begin
                    print @t1;
               end

               else if ( @i = 2 )
               begin
                    print @t2;
               end

               else
               begin
                    set @nextTerm = @t1 + @t2;
                    print @nextTerm;
                    set @t1 = @t2;
                    set @t2 = @nextTerm;
               end
```

```
            set @i = @i + 1;
        end
```

------- Output



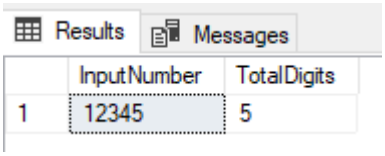

```
/*
Purpose:
--------
This script counts the total number of digits in a given number.

Examples:
---------
12345  -> 5
100    -> 3
7      -> 1

Logic:
------
1. Divide the number by 10 repeatedly.
2. Increment the counter for each iteration.
3. Continue until the number becomes zero.
*/
Declare @n int = 12345;
Declare @originalN int = @n;
Declare @count int = 0;
```

```
-- Handle 0 explicitly if necessary, otherwise the loop won't
execute
if @n = 0
    set @count = 1;
else
begin
    -- Ensure we work with absolute value to handle negative numbers
    set @n = ABS(@n);

    while @n > 0
    begin
        set @n = @n / 10;
        set @count = @count + 1;
    end
end
```

------ Query

```
select
    @originalN as InputNumber,
    @count as TotalDigits;
```

------ Output

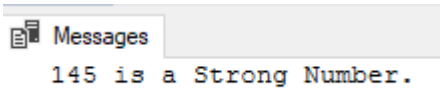



```
/*
Purpose:
--------
This script checks whether a given number is a Strong Number.

Examples:
```

```
---------
145 = 1! + 4! + 5!
    = 1 + 24 + 120
    = 145

Logic:
------
1. Extract each digit from the number.
2. Calculate the factorial of each digit.
3. Add all factorial values.
4. Compare the sum with the original number.
5. Display whether the number is Strong or Not Strong.
*/
Declare @n int = 145;
Declare @sum int = 0;
Declare @digit int;
Declare @fact int;
Declare @i int;
Declare @temp int = @n;

while @temp > 0
begin
    set @digit = @temp % 10;



    set @fact = 1;
    set @i = 1;

    while @i <= @digit
    begin
```

```
        set @fact = @fact * @i;
        set @i = @i + 1;
    end

    set @sum = @sum + @fact;
    set @temp = @temp / 10;
end

if @sum = @n
    print CAST(@n as nvarchar(10)) + ' is a Strong Number.';
else
    print CAST(@n as nvarchar(10)) + ' is not a Strong Number.';
```

-------- Output




```
/*
Purpose:
--------
This script calculates the total number of Vowels and Consonants
into string.
*/
```

```
DECLARE @String VARCHAR(100) = 'HELLO WORLD';
DECLARE @Index INT = 1;
DECLARE @Length INT = LEN(@String);
DECLARE @Char CHAR(1);

DECLARE @Vowels VARCHAR(100) = '';
DECLARE @Consonants VARCHAR(100) = '';

DECLARE @VowelCount INT = 0;
DECLARE @ConsonantCount INT = 0;
```

```
WHILE @Index <= @Length
BEGIN
    SET @Char = UPPER(SUBSTRING(@String, @Index, 1));

    IF @Char LIKE '[A-Z]'
    BEGIN
        IF @Char IN ('A', 'E', 'I', 'O', 'U')
        BEGIN
            SET @Vowels = @Vowels + @Char + ',';
            SET @VowelCount = @VowelCount + 1;
        END
        ELSE
        BEGIN
            SET @Consonants = @Consonants + @Char + ',';
            SET @ConsonantCount = @ConsonantCount + 1;
        END
    END

    SET @Index = @Index + 1;
END

-- Remove trailing comma
SET @Vowels = LEFT(@Vowels, LEN(@Vowels) - 1);
SET @Consonants = LEFT(@Consonants, LEN(@Consonants) - 1);

PRINT 'Vowels = ' + @Vowels;
PRINT 'Vowel Count = ' + CAST(@VowelCount AS VARCHAR(10));

PRINT 'Consonants = ' + @Consonants;
```

```
PRINT 'Consonant Count = ' + CAST(@ConsonantCount AS
VARCHAR(10));
```

----- Query

```
SELECT
    @Vowels AS Vowels,
    @VowelCount AS VowelCount,
    @Consonants AS Consonants,
    @ConsonantCount AS ConsonantCount;
```

----- Output

Results | Messages

	Vowels	VowelCount	Consonants	ConsonantCount
1	E,O,O	3	H,L,L,W,R,L,D	7

```
/*
Purpose:
--------
This script helps to Create Different * patterns. Used Case
statement into it. User need to add just pattern name and In output
it will automatically generate.
*/
DECLARE @Rows INT = 5;
DECLARE @PatternType VARCHAR(30) = 'Hour Glass';
DECLARE @i INT = 1;
DECLARE @j INT = 1;
DECLARE @Line NVARCHAR(100) = '';

PRINT '--- Generating Pattern: ' + @PatternType + ' ---';
IF @PatternType = 'Right Half Pyramid'
BEGIN
    WHILE (@i <= @Rows)
    BEGIN
```

```
          WHILE (@j <= @i)
          BEGIN
               SET @Line = @Line + '* ';
               SET @j = @j + 1;
          END
               PRINT @Line;
               SET @i = @i + 1;
    END
END

ELSE IF @PatternType = 'Left Half Pyramid'
BEGIN
     WHILE (@i <= @Rows)
     BEGIN
         SET @Line = '';
          SET @j = 1;

          WHILE (@j <= (@Rows - @i))
          BEGIN
               SET @Line = @Line + '  ';
               SET @j = @j + 1;
          END

              SET @j = 1;
               WHILE (@j <= @i)
               BEGIN
                    SET @Line = @Line + '* ';
                    SET @j = @j + 1;
               END
```

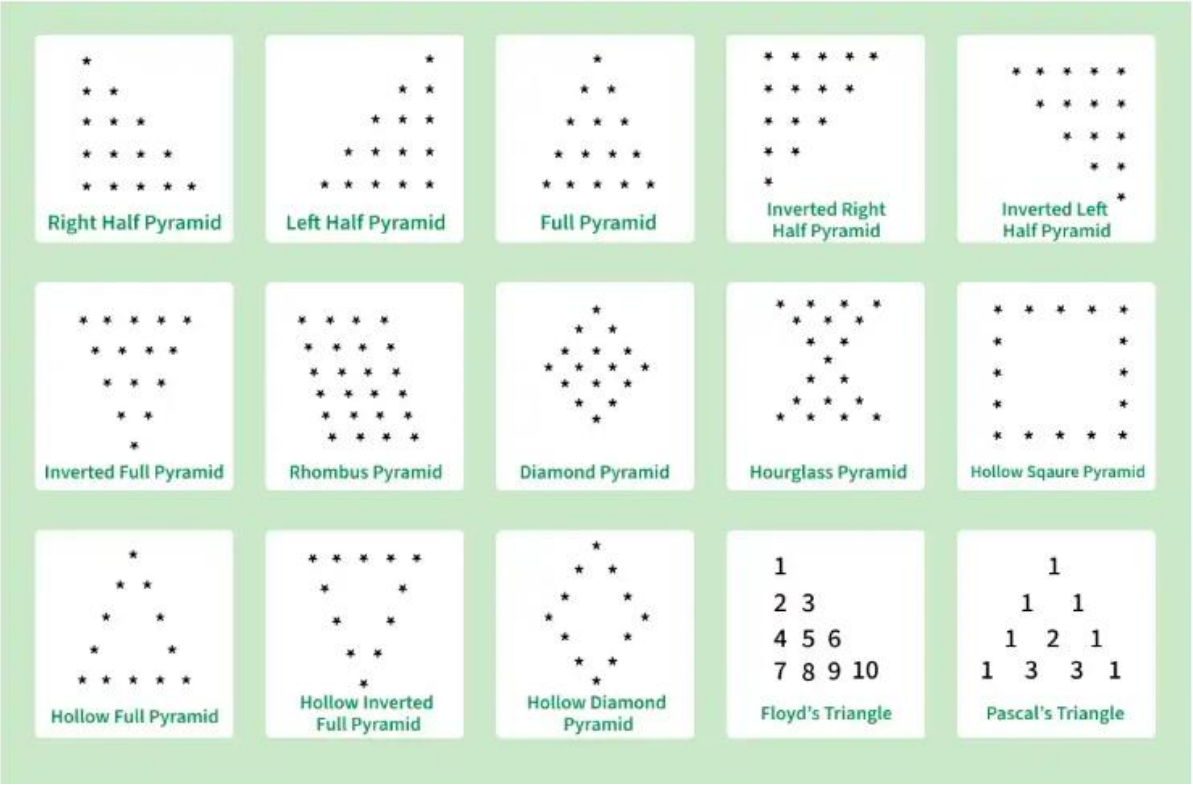
```
                PRINT @Line;
                SET @i = @i + 1;
      END
END

ELSE IF @PatternType = 'Full Pyramid'
BEGIN
      WHILE (@i <= @Rows)
      BEGIN
            SET @Line = '';

            SET @j = 1;
            WHILE (@j <= (@Rows - @i))
            BEGIN
                  SET @Line = @Line + ' ';
                  SET @j = @j + 1;
            END

            SET @j = 1;
            WHILE (@j <= (2 * @i - 1))
            BEGIN
                  SET @Line = @Line + '*';
                  SET @j = @j + 1;
            END

            PRINT @Line;
            SET @i = @i + 1;
      END
END
```

```
ELSE IF @PatternType = 'Diamond Pyramid'
BEGIN
     SET @i = 1;
     WHILE (@i <= @Rows)
     BEGIN
          SET @Line = SPACE(@Rows - @i) + REPLICATE('* ', @i);
          PRINT @Line;
          SET @i = @i + 1;
     END

     SET @i = @Rows - 1;
     WHILE (@i >= 1)
     BEGIN
          SET @Line = SPACE(@Rows - @i) + REPLICATE('* ', @i);
          PRINT @Line;
          SET @i = @i - 1;
     END
END

ELSE IF @PatternType = 'Hollow Squere'
BEGIN
    WHILE (@i <= @Rows)
    BEGIN
        SET @j = 1;
        SET @Line = '';
        WHILE (@j <= @Rows)
        BEGIN

            IF (@i = 1 OR @i = @Rows OR @j = 1 OR @j = @Rows)
```

```
                SET @Line = @Line + '* ';
            ELSE
                SET @Line = @Line + '  ';
            SET @j = @j + 1;
        END
        PRINT @Line;
        SET @i = @i + 1;
    END
END

ELSE IF @PatternType = 'Rombus'
BEGIN
     WHILE (@i <= @Rows)
     BEGIN

          SET @Line = SPACE(@Rows - @i);

          SET @Line = @Line + REPLICATE('* ', @Rows);

          PRINT @Line;
          SET @i = @i + 1;
     END
END

ELSE IF @PatternType = 'Hour Glass'
BEGIN
     SET @i = @Rows;
     WHILE (@i >= 1)
     BEGIN
          SET @Line = SPACE(@Rows - @i) + REPLICATE('* ', @i);
```

```
            PRINT @Line;
            SET @i = @i - 1;
      END

      SET @i = 2;
      WHILE (@i <= @Rows)
      BEGIN
            SET @Line = SPACE(@Rows - @i) + REPLICATE('* ', @i);
            PRINT @Line;
            SET @i = @i + 1;
      END
END

ELSE
BEGIN
    PRINT 'Invalid Pattern Type Selected';
END
```

------ Output


Right Half Pyramid
Left Half Pyramid
Full Pyramid
Inverted Right Half Pyramid
Inverted Left Half Pyramid
Inverted Full Pyramid
Rhombus Pyramid
Diamond Pyramid
Hourglass Pyramid
Hollow Sqaure Pyramid
Hollow Full Pyramid
Hollow Inverted Full Pyramid
Hollow Diamond Pyramid
1
2 3
4 5 6
7 8 9 10
Floyd's Triangle
1
1 1
1 2 1
1 3 3 1
Pascal's Triangle